

1. Title Page

Lab Name: Lab 06: TCP Connection, Flow, and Congestion Control
Date: YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

2. Background

The Transmission Control Protocol (TCP) is a core Transport Layer protocol designed to provide reliable, ordered, and error-checked delivery of data between applications. To achieve this, TCP relies on:

1. **Connection Management:** A three-way handshake (SYN, SYN-ACK, ACK) to establish connection state, and a four-step teardown (FIN, ACK) to terminate it.
2. **Reliable Data Transfer:** Using Sequence (Seq) and Acknowledgment (Ack) numbers to track byte streams, acknowledge received data, and trigger retransmissions for lost segments.
3. **Flow Control:** Using a sliding window mechanism where the receiver advertises its available buffer space (Window Size) to prevent the sender from overflowing its queue.
4. **Congestion Control:** Adjusting the sending rate dynamically (slow start and congestion avoidance) based on perceived network capacity, packet loss, and round-trip time (RTT).

In this lab, you will generate high-volume TCP traffic across a constrained routing topology in CML, analyze the packets in Wireshark, and plot TCP congestion graphs.

3. Objective / Purpose

To analyze connection establishment/termination, trace reliable data transfer (Seq/Ack tracking), examine flow control advertising (Window Size), calculate throughput, and graph TCP congestion control behavior under constrained link conditions.

4. Topology Diagram

```
graph LR
    H0["Host H0 (TCP Client)<br>IP: 10.0.1.10/24<br>Gateway: 10.0.1.1"] <--> |"eth0 <--> Gi0/0"| R0["Router R0 (IOSv)<br>Gi0/0: 10.0.1.1<br>Gi0/1: 10.0.2.1"]
    R0 <--> |"Gi0/1 <--> eth0<br>[Constrained Link / Link Condition]"| H1["Host H1 (TCP Server)<br>IP: 10.0.2.10/24<br>Gateway: 10.0.2.1"]

    classDef node fill:#d4f1f9,stroke:#007b8b,stroke-width:2px;
    classDef router fill:#fff2cc,stroke:#d6b656,stroke-width:2px;
```

```
class H0,H1 node;
class R0 router;
```

Details: - 2x Hosts (<initials>-H0 and <initials>-H1) communicating over a routed CML topology via Router (<initials>-R0). - Link constraints (bandwidth limits and latency) are applied to the link between R0 and H1 to force congestion control behavior.

5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 2x Linux Nodes (configured as hosts), 1x IOSv Router

6. Configuration / Methodology

IP Addressing Table

Device	CML Node Name	Interface	IP Address	Subnet Mask	Gateway
Client	<initials>-H0	eth0	10.0.1.10	255.255.255.0	10.0.1.1
Server	<initials>-H1	eth0	10.0.2.10	255.255.255.0	10.0.2.1
Router	<initials>-R0	G0/0	10.0.1.1	255.255.255.0	N/A
Router	<initials>-R0	G0/1	10.0.2.1	255.255.255.0	N/A

CML Environment Setup

1. Verify that your hosts and router <initials>-R0 are running and have the correct interfaces and IP settings. Refer to Lab 5 if you need to reconfigure <initials>-R0.
2. Configure **Link Condition** constraints on the link between **Router R0 and Host H1**:
 - **Bandwidth:** Limit the link to **1 Mbps** (or 1000 Kbps).
 - **Latency:** Set a one-way delay of **50 ms** (100 ms RTT).
 - *Note: These constraints will limit link capacity, ensuring TCP enters congestion avoidance and slow-start cycles during file transfers.*

Step-by-Step Traffic Generation & Execution:

1. **Start Packet Capture:** Right-click the link connecting **Host H0 and Router R0**, select **Packet Capture**, and click **Start**.

2. **Generate 150KB File on Host H0:** Open H0's console and create a dummy text file of exactly 150KB:

```
dd if=/dev/urandom of=alice.txt bs=1k count=150
```

3. **Start TCP Listener on Host H1:** Open H1's console and start netcat in TCP listen mode, discarding the incoming stream:

```
nc -l -p 8080 > /dev/null
```

4. **Send the File from Host H0:** Open H0's console and pipe the file through netcat to H1:

```
nc 10.0.2.10 8080 < alice.txt
```

5. **Wait and Terminate:** Once the file transmission completes (the console prompt will return on H0), verify that the connection has completed. Press Ctrl+C on H0 to close netcat if it remains open.
 6. **Download Capture:** Stop the CML packet capture and download the .pcap capture file to your local computer.
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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark, filter the packets using the tcp filter, and answer the following questions:

Part A: TCP Basics and Handshake

1. **Client & Server Details:** What is the IP address and TCP port number used by the client computer (<initials>-H0) to transfer the file? What is the IP address and TCP port number used by <initials>-H1?
2. **SYN Segment:** Locate the TCP SYN segment that initiated the connection.
 - What is the raw Sequence Number of this segment? (Note: To see the raw value, right-click on the sequence number in the packet details pane -> Protocol Preferences -> uncheck Relative Sequence Numbers).
 - What flags are set in the TCP header to identify this as a SYN packet?
 - Does this segment advertise support for Selective Acknowledgments (SACK) in its Options?
3. **SYN-ACK Segment:** Locate the SYN-ACK segment sent by the server.
 - What is its Sequence Number?
 - What is the value of the Acknowledgment number field? How did the server determine this value?
 - What flags are set in the TCP header?
4. **Connection Teardown:** Locate the connection closing packets (FIN/ACK sequence).
 - What is the packet number of the first segment that initiated the teardown (FIN)? Which host sent it?
 - Outline the packet sequence exchanged to close the connection.

Part B: Reliable Data Transfer

5. **First Data Segment:** Locate the first TCP segment containing the HTTP/netcat payload sent from H0 to H1.
 - What is the Sequence Number of this segment?
 - How many bytes of data are contained in the TCP payload (data) field of this segment?
6. **Data Acknowledgment:** Locate the ACK segment sent by the server acknowledging this first data segment.
 - What is the Sequence Number of this ACK segment?
 - What is the Acknowledgment number value? How did the server determine this value?
7. **Retransmission Check:** Are there any retransmitted segments in the trace file? What did you check in Wireshark (filters or packet fields) to answer this?

Part C: Flow Control and Throughput

8. **Segment Lengths and Window Size:** Examine the first four data-carrying TCP segments sent from H0 to H1.
 - What is the length (header plus payload) of each of these segments?
 - What is the minimum amount of available buffer space advertised to the client by <initials>-H1 among these first four segments? (Inspect the Window Size Value field).
 - Did the lack of receiver buffer space ever throttle the sender during these segments?
9. **Delayed ACKs:** How much data does the receiver typically acknowledge in a single ACK among the first ten data-carrying segments? Can you identify cases where the receiver is ACKing every other received segment (delayed ACKs)?
10. **Throughput Calculation:** What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value (indicate the start/end timestamps and total payload bytes).

Part D: Congestion Control in Action

11. **Stevens Graph Plot:** Click on a client-sent TCP segment in the Wireshark packet list. Then select: Statistics -> TCP Stream Graph -> Time Sequence Graph (Stevens). > **IMPORTANT - Deliverable Screenshot:** Take a screenshot of the generated Stevens graph. The X-axis represents time, and the Y-axis represents TCP sequence numbers. Include this graph in your report.
 12. **Congestion Phases Analysis:** Analyze the Stevens graph:
 - Describe the pattern of the dots (e.g., clusters/fleets of packets sent back-to-back, followed by flat periods).
 - Can you identify where the connection is in the **slow start** phase and where it transitions to the **congestion avoidance** phase? Explain what indicators on the graph support your analysis.
 13. **RTT Graphing (Optional / Extra Credit):** Plot the RTT for the TCP segments sent from the client. Select: Statistics -> TCP Stream Graph -> Round Trip Time Graph.
 - What is the average RTT observed? Does this match the Link Condition latency of 100ms round-trip time?
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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything]

Discovery Method: [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

9. Conclusion & Analysis

Summarize your findings regarding TCP's connection management, reliability mechanisms, and how the sliding window flow control and congestion avoidance phases ensure network stability.

10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
 - Your written answers to the 13 Wireshark analysis questions in Section 7.
 - A screenshot of your active CML topology showing the running hosts (<initials>-H0, <initials>-H1) and router (<initials>-R0).
 - An annotated screenshot of your Wireshark packet list showing the TCP 3-way handshake (SYN, SYN-ACK, ACK).
 - An annotated screenshot of your Wireshark packet list showing the connection teardown (FIN, ACK, FIN, ACK).
 - The required **Time Sequence Graph (Stevens)** screenshot generated in Wireshark as detailed in Part D.